Module Guide

Site: Eduvos LMS
Course: Machine Learning Algorithms
Book: Module Guide
Printed by: Sekwanele Njoko
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1. Module Information

Faculty	Information Technology	NQF Level	6
Module Code	ITMLA2	Credit Value	12
Module Name	Machine Learning Algorithms	Notional Hours	120
Year Level	2	Prerequisites/ Corequisites	ITSCA2 (P)
Duration of Module	1 Block	Copy Editor	Mr Kyle Keens

The module guide must be read in conjunction with the textbook and study manual. This document will be the first port of call to understanding what will be assessed and which assessments form part of the module.

The purpose of the module guide is to highlight:

- The learning outcomes and assessment criteria that need to be met to pass the module
- The assessments required to be completed for the module
- The additional resources required for the module
- The topics that will be focused on for the module

2. Module Aim

This module aims to impart students with a deep understanding of the mathematical principles behind machine learning algorithms and practical skills for applying them to real-world case studies in areas such as data science, robotics, and computer science, facilitating their practical use in these fields.

3. Module Description

The main objective of machine learning is to assist computer systems to discover unique patterns in specific data which then allows the system to make predictions based on the pattern discovered. These often-complex patterns, serve as the basis for answering business questions and solving problems. Machine learning is a type of artificial intelligence (AI) that provides computers with the ability to learn without being explicitly programmed. Machine learning finds patterns in large volumes of data and uses those patterns to perform predictive analysis.

In this module, supervised and unsupervised machine learning (ML) approaches are discussed including complex searches using metaheuristic methods to cover four main intelligent algorithms families: regression analysis, classification, clustering analysis as well as complex searches. Throughout the course, students will be exposed to various types of ML algorithms such as Linear regression, Logistic regression, Naïve Bayes, Decision Trees, artificial neural networks, k-Means clustering and genetic algorithms.

4. Learning Outcomes

By the end of this module, you should be able to:

Learning Outcome	Assessment Criteria

1. Describe the theoretical basics of machine.	1.1 Describe data in machine learning. 1.2 Discuss machine learning algorithm principles for predictive modelling. 1.3 Differentiate between parametric and non-parametric machine learning algorithms. 1.4 Discuss the difference between machine learning algorithms. 1.5 Describe the bias-variance trade-off. 1.6 Explain the implications of overfitting and underfitting.
2. Implement the mathematical concepts of linear machine learning algorithms.	2.1 Explain the underlying principle of linear machine learning techniques. 2.2 Demonstrate the use of gradient descent in machine learning. 2.3 Discuss the linear regression algorithm for real-valued predictions. 2.4 Discuss the logistic regression algorithm for binary classification. 2.5 Implement linear and logistic regression algorithms using stochastic gradient descent. 2.6 Describe the pertinence of feature selection methods in machine learning. 2.7 Demonstrate proficiency in commonly used effective feature selection methods.
3. Apply the mathematical concepts of non-linear machine learning algorithms.	3.1 Explain the underlying concept behind nonlinear machine learning algorithms. 3.2 Discuss and implement the Classification and Regression Trees (CART) models for data prediction. 3.3 Discuss and implement Naive Bayes algorithms for categorical variables. 3.4 Implement the learning architecture of artificial neural networks. 3.5 Show how to use feedforward neural networks as regression algorithms. 3.6 Implement feedforward neural networks as classifiers.
4. Implement cluster analysis.	4.1 Discuss the principle and benefits of clustering analysis as an unsupervised machine-learning technique. 4.2 Explain the underlying principle of k-Means clustering. 4.3 Implement k-Means clustering on unlabeled datasets. 4.4 Implement optimal clusters selection approach for effective k-Means clustering.

5. Demonstrate the concept of complex decision-making using metaheuristic algorithms.	5.1 Apply the concept of complex decision-making using computer-based algorithms. 5.2 Analyse the underlying principle of genetic algorithms. 5.3 Implement genetic algorithms within a computing environment. 5.4 Solve real-world problems using genetic algorithms.

5. myLMS Module Page

This myLMS page is built to ensure that you receive guidance on how to approach your module and it includes documentation to assist you in succeeding. You will be required to engage **daily** with all the material on your myLMS page to enable you to pass the module. It is important that you utilise the following resources available on your myLMS module page:

- Module guide
- Module lessons
- Blended learning activities
- Assessments
- Important notifications from your lecturer
- Module announcements (discussion forums)
- Links to lectures, lecture recordings and prescribed videos (where applicable)

Additional academic documents can be found on the myDocuments myLMS page.

6. Prescribed Resources

Textbook(s)

M. Gopal, Applied Machine Learning, McGraw-Hill Education India. 2nd edition. 2021. Print ISBN: 9789354601590, 9354601596, eText ISBN: 9789354601408, 9354601405

7. Recommended Resources

Textbook(s)

Bishop, M. 2006. *Pattern Recognition and Machine Learning*: Springer

Theodoridis, S. 2008. *Pattern Recognition*: Academic Press

Journal Article(s)

Imam. F, Musilek. P, Reformat. M. Parametric and Nonparametric Machine Learning Techniques for Increasing Power System Reliability: A Review. Information, 2024

Ezugwu. E, Ikotun. M, Oyelade. O, Abualigah. L, Agushaka. O, Eke. C, Akinyelu. A. comprehensive survey of clustering algorithms: State-of-the-art machine learning applications, taxonomy, challenges, and future research prospects. Engineering Applications of Artificial Intelligence, 2022.

Meena. K and Suriya. S. A Survey on Supervised and Unsupervised Learning Techniques, pages 627–644. 2020.

Video(s)

MIT OpenCourseWare. 2010. *MIT 6.034 Artificial Intelligence [Video]*. YouTube.

. [Last accessed:10/12/2024]

Standford Online. 2018. *Standford CS229: Machine Learning Full Course taught by Andrew Ng [Video]*. Youtube. https://youtube.com/playlist?list=PLoROMvodv4rMiGQp3WXShtMGgzqpfVfbU&si=5wmuh50E8TWPlRtE. [Last accessed:10/12/2024]

Sentdex. 2017. *Machine Learning with Python [Video]*. YouTube.

. [Last accessed:10/12/2024]

8. Supporting Documents

Geyer, L., Levin, A., Makati, P., Pierce, R., Potter, M., and Wheeler, A. 2019. *Eduvos Guide to Referencing (Harvard Referencing Method)*. Unpublished document. Eduvos.

9. Essential Requirements

- Access to the institution's Learning Management System (myLMS) to access all study material.
- Access to Microsoft Teams to attend online lectures and consultations.
- Access to a resource centre or an online library with a wide range of relevant resources including textbooks, newspaper articles, journal articles, organisational publications and databases.
- Access to a range of academic journals in electronic format via ProQuest or other databases.

10. Employer Engagement and Vocational Contexts

- Work-integrated learning will be built in to involve using machine learning algorithm concepts for the purpose of applying such knowledge in data prediction and analysis.
- Students should use what they have learned during this course to program and implement different types of machine-learning algorithm concepts relevant to the field of IT.

11. Time Allocation Guideline: Full-Time Students

	Week 0	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Lecturer-led Sessions	No live sessions	4	4	4	4	4	4	4	0
Practicals / Tutorials	No sessions	1	1	1	1	1	1	1	0
myLMS Lessons and Activities	Learning starts	6	6	6	6	6	6	7	0
Formative Assessments[1]	0	6.5	6.5	6.5	6.5	6.5	6.5	0	0
Summative Assessments[2]	0	0	0	0	0	0	0	0	3
Total	0	17.5	17.5	17.5	17.5	17.5	17.5	12	3
Overall Total	120 Hours								

[1] This time includes the preparation time and assessment time.

[2] This time includes the preparation time and assessment time.

12. Time Allocation Guideline: Part-Time Students

	Week 0	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8
Lecturer-led Sessions	No live sessions	4	4	4	4	4	4	4	0
Practicals / Tutorials	No sessions	1	1	1	1	1	1	1	0
myLMS Lessons and Activities	Learning starts	6	6	6	6	6	6	7	0
Formative Assessments[1]	0	6.5	6.5	6.5	6.5	6.5	6.5	0	0
Summative Assessments[2]	0	0	0	0	0	0	0	0	3
Total	0	17.5	17.5	17.5	17.5	17.5	17.5	12	3
Overall Total	120 Hours								

[1] This time includes the preparation time and assessment time.

[2] This time includes the preparation time and assessment time.

13. ICT Requirements

ICT Required	Reason	Lecture Week(s)
Hardware Requirements: Minimum: Intel Core™ i5 8th Generation or higher / AMD Ryzen™ 5 3rd Generation or higher Trusted Platform Module (TPM) version 2.0 8GB RAM 256GB SATA SSD Recommended (considering certain software and hardware-based modules): Intel Core™ i5 8th Generation or higher / AMD Ryzen™ 5 3rd Generation or higher Trusted Platform Module (TPM) version 2.0 16GB RAM 512GB SATA / NVMe SSD Optimal: Intel Core™ i7 8th Generation or higher / AMD Ryzen™ 7 3rd Generation or higher Trusted Platform Module (TPM) version 2.0 32GB RAM 512GB NVMe SSD	Lab work, Lectures, Assessments	1-8
Software Requirements: Computer with • Anaconda Spyder • Python 3.0 • Python Numpy • Python Scipy • Python Matplotlib • Python Scikit-learn	Lab work, Lectures, Assessments	1-8

14. Assessment

Assessments will take place continually during the block to evaluate your progress throughout the module.

Assessment = Formative assessment + Summative assessment

All assessments must be submitted through Turnitin, on the assessment tile, within your myLMS module page. **Assessments must not be submitted via Turnitin on the campus page(s).**

You may be required to complete some assessments online on myLMS. The following guidelines may apply:

- You may be required to write an online assessment directly on myLMS. You will need to log in at a specified time to attempt the assessment. Once the allocated time expires, the assessment will no longer be available to complete.
- Time limits should be checked before commencing assessments. Refer to the Details of Assessments table in this module guide.
- Ensure that your internet connection is stable.

- In some cases, assessments will not be available indefinitely and will be available for only a day or two.
- The marks for some assessments may only be available after all students have attempted the assessment after the assessment due date.

It is the students' responsibility to retrieve their assessment feedback and verify their marks on the day they are released. No adjustment of marks will be entertained beyond one week after marks are released.

15. Student Guidance on AI Use

This module follows an academic quality–based approach to assessment. This means that your work will be evaluated primarily on the quality of your thinking, analysis and academic engagement, rather than on whether AI tools were used.

You are expected to:

- Submit work that demonstrates your own understanding, reasoning and disciplinary knowledge.
- Ensure that any use of AI (where permitted) does not replace your learning or independent engagement.
- Take full responsibility for the accuracy, originality and academic quality of your submission.

AI use in this module is governed at the level of each assessment. Each task will specify its AI-use category, which outlines:

- Whether AI tools are permitted.
- The extent to which they may be used.
- Any declaration or process evidence requirements.

To understand what is expected of you in a specific assessment:

- Refer to the "Details of Assessments" table in your module guide.
- Each assessment will further stipulate the AI-use category and associated requirements.

Where AI use is permitted, you must follow the instructions provided and complete any required AI disclosure appendix. Where AI use is restricted or not allowed, you must ensure that your work reflects your own unaided effort.

Failure to comply with the stated AI requirements for an assessment may impact your marks or be treated as a matter of academic integrity.

AI Use Categories Table

Code	Category	What This Means	Evidence Required	Disclosure Required
AI-0	**No AI use**	You are not allowed to use any AI tools at any stage of this assessment. All work must be entirely your own.	No additional evidence is required. Your submission should reflect your own independent work.	No disclosure appendix is needed.
AI-1	**Limited AI use**	You may use AI only for simple support tasks (e.g. checking grammar or generating initial ideas). The main work (analysis, arguments, writing) must still be your own.	You must complete and submit an AI disclosure appendix explaining how you used AI.	You must complete and submit an AI disclosure appendix explaining how you used AI available in the assessment template on the myDocuments page.
AI-2	**Permitted with declaration**	You may use AI tools as an aid to support your work. However, you remain responsible for the accuracy, quality and originality of your submission.	You must complete and submit an AI disclosure appendix explaining how you used AI.	You must complete and submit an AI disclosure appendix explaining how you used AI available in the assessment template on the myDocuments page.
AI-3	**Integrated / Exploratory**	AI is a core part of the task. You are expected to use AI, but also to critically evaluate its outputs and show how it contributed to your work.	You must complete and submit an AI disclosure appendix explaining how you used AI..	You must complete and submit an AI disclosure appendix explaining how you used AI available in the assessment template on the myDocuments page.

16. Formative Assessment

16.1. Project

This module comprises one project to be completed individually. The project contributes 40% towards the student's final mark.

Projects must be submitted on or before the due date to the lecturer as per arrangement (or as otherwise stipulated). Five per cent (5%) will be deducted for every day that the Project is late, up to a maximum of three days. Projects that are more than three days late will be awarded a zero.

16.2. Online Test

This module comprises one online test which contributes 10% towards the student's final mark.

If an assessment is missed because of illness, a doctor’s note must be presented within 3 calendar days of the missed assessment to the academic admin manager/administrator/ coordinator.

To make up for this missed assessment, you may be able to attempt a deferred assessment. However, in order to gain entry to this assessment, you will have to follow various procedures and meet certain criteria. You must complete a Deferred Assessment Application Form available on myLMS. You will be required to pay a non-refundable fee per application. Each assessment missed requires a separate application. This will be your only opportunity to make up for a missed assessment.

17. Summative Assessment

The pass mark for a summative assessment is 50%. In order to pass the module, a sub-minimum mark of 40% or higher is required for the summative assessment and a final average of 50% or higher is required for the entire module.

A summative assessment schedule will be released to students on myLMS with their timetabled assessment dates before Week 5 of each block. This schedule will be made available on the MyDocuments page on myLMS.

17.1. Proctored Online Assessment

There will be one **Proctored** Online Assessment to complete in the summative assessment period that will count 50% towards your final mark.

The summative assessment has a sub-minimum requirement of at least 40%. Supplementary opportunities may be available for selected assessments (refer to the Details of Assessments table) for students who fail the summative assessment(s), and these will cover the entire module’s content. These will be scheduled in the allocated weeks as per the academic year calendar and may be in the form of a re-submission opportunity dependent on the type of summative assessment.

To make up for a missed summative assessment, you may be able to attempt a deferred summative assessment. This is dependent on the circumstance that affected your ability to attempt the original assessment. However, to qualify for the deferred assessment, you will have to follow various procedures and meet certain criteria. You must complete a *Deferred Assessment Application Form* available on myLMS, and you will be required to pay a non-refundable fee per application. The deferred summative assessment will be attempted at the same time as the supplementary summative assessment and will cover the entire

module's content. A student who has been granted a deferred assessment and who does not present themselves, or fails the assessment, will not be permitted a supplementary opportunity, and will need to re-attempt the module as stipulated in the Conditions of Enrolment.

18. Plagiarism

All assignments and reports must be submitted to the online similarity checker (Turnitin) available on myLMS prior to being submitted for marking. When submitting your assignment/report, it is compulsory to submit the entire Turnitin report. Marks will be deducted in accordance with the institutional policy.

Also, when submitting assessments you should include the applicable completed and signed assessment coversheet as an acknowledgement that the work submitted is your own original work, except for source material explicitly acknowledged. This declaration will serve as proof that you are aware of the institution's policies and regulations on academic integrity.

19. Final Mark

To pass the module, a final average of 50% or higher is required for the entire module.

The final mark is calculated as follows:

Final Mark = [(Project percentage × 0.40) + (Online test percentage × 0.10) + (**Proctored** Summative assessment percentage × 0.50)]

20. Compiling a Portfolio of Evidence

You must demonstrate, through the presentation of evidence, that you have met all module requirements within the qualification being undertaken. To do this, you must organise your evidence into what is known as a 'portfolio'.

A portfolio will take time and effort to complete. It is a means of focusing and demonstrating to others your strengths and achievements. A portfolio is an important resource that you may find useful to retain once you have achieved your qualification, particularly when applying for future positions.

You are encouraged to read more about building a portfolio and to begin populating your evidence to illustrate your full skill set to future employers.

21. Details of Assessments

Assessment	Assessment Method	AI-Category	Weighting	Lecture Week	Sub-Minimum Requirement	Release Week	Submission Week	Submission Time	Duration	Proctoring Requirements
Formative Assessment: Online Test (Covers Weeks 1 – 3)	Online Test	AI-0	10%	Week 4	N/A	Tue, Week 4	Mon, Week 5	23:59	90 Minutes	N/A
Deferred Formative Assessment: Online Test (Covers Weeks 1 – 3)	Deferred Online Test	AI-0	10%	Week 6	N/A	Wed, Week 6	Mon, Week 7	23:59	90 Minutes	N/A
Formative Assessment: Project (Covers Weeks 1– 6)	Project	AI-1	40%	Week 6	N/A	Week 1	Week 6	23:59	N/A	N/A
TOTAL WEIGHT OF FORMATIVE ASSESSMENTS			50%							

Summative Assessment (covers all Learning Outcomes)	Online Assessment	AI-0	50%	Week 8	40%	As per summative assessment schedule	As per summative assessment schedule	As per summative assessment schedule	3 Hours	Invigilator App
Supplementary/Deferred Summative Assessment (covers all Learning Outcomes)	Online Assessment	AI-0	50%	As per summative assessment schedule	40%	As per summative assessment schedule	As per summative assessment schedule	As per summative assessment schedule	3 Hours	Invigilator App
TOTAL WEIGHT OF SUMMATIVE ASSESSMENTS			50%							
TOTAL WEIGHT			100%							

***Note:** Check the AI category for each assessment in this module. Refer to the *Student Guidance on AI Use* in your module guide, and review the assignment instructions and assessment brief on your assessment page for specific AI requirements.

Proctoring Requirements Key

CO = **On-campus online assessment** in an invigilated PC venue. If you want to attempt at a campus that is not your preferred campus a seat must be booked at least 2 days prior to the assessment.

CP = **On-campus paper-based** assessment in an invigilated venue. If you want to attempt at a campus that is not your preferred campus a seat must be booked at least 2 days prior to the assessment.

IA = Off-campus online assessment: **The Invigilator** proctoring software enabled. If you want to attempt at a campus a seat must be booked for an invigilated venue at least 2 days prior to the assessment.

R = Off-campus online assessment: **Respondus** proctoring software enabled. If you want to attempt at a campus a seat must be booked for an invigilated venue at least 2 days prior to the assessment.

22. Consultations

Lecturers will be available for consultations. Specific details in this regard will be made available on your campus specific myLMS pages. You must give lecturers 24 hours' notice for appointments. Consultation meetings can be requested via email. It is important that you detail the requirements (chapter, section, etc.) for your consultation.

23. Module Content

You are required to attend all lessons. In addition, exercises and activities, which are supplied by lecturers, are compulsory.

23.1. First Block

Learning Weeks	Topics and Assessment Criteria Covered	Assessments	References
Week 0	Induction and Orientation	**Baseline Assessment**	MyLMS
Week 1	Theoretical basics of machine learning AC: 1.1 - 1.6	**Project released**	• Chapter 1,2 ◦ Section 1.1-1.9,2.1-2.3 ◦ MyMLS page Week 1

Week 2	Linear machine learning: Gradient descent, linear regression and logistic regression AC: 2.1 - 2.5		• Chapter ◦ Section 3.6-3.7, 2.6-2.8 ◦ MyMLS page Week 2
Week 3	Feature selection methods and classification with logistic regression AC: 2.4- 2.7		• Chapter 3,2 ◦ Section 3.7, 2.8 ◦ MyMLS page Week 3
Week 4	Nonlinear machine learning techniques: Naive Bayes and Regression Trees models. AC: 3.1 - 3.3	**Online Test**	• Chapter 3, 8 ◦ Section 3.1-3.3, 8.1-8.6 ◦ MyMLS page Week 4
Week 5	Classification and regression with feedforward neural network. AC: 3.4 - 3.6		• Chapter 5 ◦ Section 5.1-5.7 ◦ MyMLS page Week 5
Week 6	Unsupervised learning with K-Means clustering algorithm. AC: 4.1 - 4.4	**Project Submission** **Deferred Online Test**	• Chapter 7 ◦ Section 7.1-7.4 ◦ MyMLS page Week 6
Week 7	Complex searches using Genetic Algorithms AC: 5.1-5.4		• Appendix A ◦ MyMLS page Week 7
Week 8	**Summative Assessment Week**		
	Supplementary/Deferred Assessment		